Chiye Ma

Ph.D. Candidate in Mechanical Engineering

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Summary

Ph.D. candidate in Mechanical Engineering at City University of Hong Kong. Research focuses on manufacturing systems, supply chain management, stochastic modeling, optimization, and reinforcement learning for operational decision-making under uncertainty.

Education

Present Sep. 2023	Mechanical Engineering, **City University of Hong Kong** Ph.D. Candidate in Mechanical Engineering
Jun. 2023 Sep. 2020	School of Automation, **Beijing Institute of Technology** M.Eng. in Control Engineering
Jun. 2020 Sep. 2016	Electronic Information and Automation, **Tianjin University of Science & Technology** B.Eng. in Measurement and Control Technology & Instrument

Research Interests

- Operations and supply chain analytics
- Stochastic manufacturing and service systems
- Learning-based decision-making under uncertainty

Publications

Published Journal Papers

1 **C. Ma**, Z. Jia, J. Zhang, M. Duan, and W. Wang, "Dynamic Modeling and Analysis of Bernoulli Production Lines With Small-Lot Orders and Product Quality Problems," *IEEE Transactions on Industrial Informatics*, 2024.

2 S. Chen, **C. Ma**, W. Wang, and E. Zio, "An agent-based cooperative co-evolutionary framework for optimizing the production planning of energy supply chains under uncertainty scenarios," *International Journal of Production Economics*, 2024.

3 Z. Jia, Z. Ni, and **C. Ma**, "Dynamic Process Modeling and Real-Time Performance Evaluation of Rework Production System With Small-Lot Order," *IEEE Robotics and Automation Letters (RA-L)*, 2023.

4 J. Zhang, E. Zio, **C. Ma**, K. Liu, and W. Wang, "A probabilistic cost-benefit analysis approach for cyberattack path evaluation," *Reliability Engineering & System Safety*, 2025.

Manuscripts

5 **C. Ma**, T. Yu, H. Chen, X. Pan, J. Zhao, and W. Wang, "A transfer learning-assisted unsupervised clustering approach for unknown fault diagnosis of nuclear power plant equipment," *Progress in Nuclear Energy*, under second-round review.

6 **C. Ma**, S. Chen, E. Zio, J. Chen, Z. Jia, and W. Wang, "SChaLM: A lightweight domain-specific language model for real-time response to disruption in industrial supply chains," submitted to *Reliability Engineering & System Safety*.

7 **C. Ma**, W. Wang, and M. Xie, "Risk-Averse Multi-Agent Reinforcement Learning for Inventory Management in Single-Manufacturer Multi-Retailer Systems Under Uncertain Disruptions," manuscript in preparation; target journal: *INFORMS Journal on Computing*.

Projects & Patents

- Research contributor to three funded projects on production-system performance, cyber-physical risk, and autonomous production-line decision-making.
- Co-inventor of two granted Chinese patents: CN114491994B and CN114398782B.

Skills

Methods	Stochastic modeling, optimization, reinforcement learning, multi-agent systems
Programming	Python, MATLAB, C
Tools	PyTorch, Pandas, Matplotlib, Git, SSH, LaTeX, Linux
Languages	Chinese (native), English (IELTS 7.0, C1)